Problem Statement

Personal Knowledge Base Assistant — AI-Powered RAG System

Project Name	Personal Knowledge Base Assistant
AI Model	Gemini 2.5 Flash
Architecture	Retrieval-Augmented Generation (RAG)
Document Types	PDF, Word, Text/Markdown, Web URLs
Document Version	1.0

1. Project Overview

In the modern information age, individuals and professionals accumulate vast quantities of documents, notes, research papers, and web content across disparate sources. Retrieving specific information from this growing corpus is time-consuming, error-prone, and cognitively demanding. There is a critical need for an intelligent system that can understand, index, and answer questions from a user's personal collection of documents in a grounded, reliable, and professional manner.

2. Problem Statement

Users who manage large personal or professional knowledge bases face the following core challenges:

- **Information Overload:** Documents accumulate across PDFs, Word files, notes, and web pages with no unified way to query them.
- **Manual Search Inefficiency:** Keyword-based search fails to understand context, intent, or relationships across documents.
- **Knowledge Fragmentation:** Related information is scattered across multiple files and formats, making synthesis difficult.
- **Lack of Grounded AI Answers:** General AI models answer from training data, not the user's actual documents, leading to hallucinations and irrelevant responses.
- **No Cross-Document Reasoning:** Users cannot easily compare, contrast, or extract insights across two or more documents simultaneously.

3. Proposed Solution

The Personal Knowledge Base Assistant is an AI-powered RAG (Retrieval-Augmented Generation) system that allows users to upload their documents and interact with them through natural language queries. The system retrieves the most relevant content chunks from a vector database and passes them to Gemini 2.5 Flash, which generates precise, cited, and grounded answers — never hallucinating beyond the provided knowledge base.

4. Core Capabilities

Question & Answer	Answer specific queries using only content from the user's knowledge base, with full source citations.
Document Summarization	Generate structured summaries of individual documents with key topics and takeaways.
Key Insight Extraction	Identify and present the most important facts, figures, and decisions from any document.
Document Comparison	Analyse and compare two or more documents side-by-side, highlighting agreements and contradictions.
Follow-up Conversations	Maintain session context to support multi-turn dialogue and progressive exploration of the knowledge base.

5. Technical Architecture

➜ **Step 1 — Ingestion:** User uploads PDF, Word, Markdown, or URL. Documents are parsed, cleaned, and split into chunks.

➜ **Step 2 — Embedding:** Each chunk is converted into a vector embedding using a text embedding model.

➜ **Step 3 — Indexing:** Embeddings are stored in a vector database (e.g. ChromaDB, Pinecone, FAISS).

➜ **Step 4 — Retrieval:** At query time, the user's question is embedded and top-K similar chunks are retrieved.

➜ **Step 5 — Generation:** Retrieved chunks + user query are passed to Gemini 2.5 Flash via the master system prompt.

➜ **Step 6 — Response:** Gemini returns a grounded, cited answer based solely on the retrieved context.

6. Supported Document Types

Format	Extension	Use Case
PDF	.pdf	Research papers, reports, scanned documents, books
Word Document	.docx	Memos, contracts, essays, meeting notes
Text / Markdown	.txt / .md	Personal notes, wikis, technical documentation
Web Pages	URL	Online articles, blog posts, online references

7. Expected Outcomes

✓ Users can query any document in their knowledge base using natural language.

✓ All answers are grounded exclusively in user-provided documents — zero hallucination.

✓ Every response includes source citations with document name and section reference.

✓ Users can compare multiple documents and receive structured, side-by-side analysis.

✓ The system maintains conversation context to support in-depth exploration across turns.

✓ A confidence level (High / Medium / Low) is attached to every response for transparency.

8. Constraints & Limitations

■ The assistant will only answer from uploaded documents. It will not use external or training knowledge.

■ Documents must be uploaded and indexed before they can be queried.

■ Accuracy of retrieval depends on the quality of embeddings and chunking strategy.

■ Very large documents may require chunking optimisation to fit within context limits.

■ Scanned PDFs without OCR pre-processing may yield incomplete text extraction.